

GreenLeaf Markets

Sales Performance Analysis Report

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Tools Used: Microsoft Excel, SQL, Tableau

Course: Data Analytics Essentials-Cisco Networking Academy

Executive Summary

This report presents a detailed analysis of GreenLeaf Market Sales dataset spanning 2023 to 2026 using Microsoft Excel for data cleaning and analysis, SQL for database querying and Tableau for interactive dashboard visualization in order to identify profitability issues and provide actionable business recommendations.

Key finding reveals that Chairs generates the highest sales volume of \$335,768 but maintains a dangerously low profit margin of 4.71% making them highly vulnerable to discounting. Despite showing positive total profit the Central region maintains a negative average profit margin of -10% indicating that widespread unprofitable transactions are masked by a few high value orders. Furthermore, analysis confirms that higher discounts consistently lead to lower profit margins with Binders recording the highest average discount and the lowest profit margin confirming that aggressive discounting is the biggest driver of unprofitable transactions in the business.

Based on these findings three strategic recommendations are provided – reducing discounts on Binders to prevent further loss in profit, conducting a price audit of the central region to address widespread losses and investing in the East region's consistent growth to maximize future revenue potential.

Data Understanding

The dataset used in this analysis is the GreenLeaf Markets Sales Datasets containing 10,194 rows and 21 columns spanning retail sales transactions from 2023-2026. It includes information on orders, customers, products, regions, sales, profit and discounts across four regions– West, East, Central and South.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	State/Province	Postal Code	Region
1	US-2023-103800	03/01/2023	01/07/2023	Standard Class	DP-13000	Darren Powers	Consumer	United States	Houston	Texas	77095	Central
2	US-2023-112326	04/01/2023	01/08/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	Illinois	60540	Central
3	US-2023-112326	04/01/2023	01/08/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	Illinois	60540	Central
4	US-2023-112326	04/01/2023	01/08/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	Illinois	60540	Central
5	US-2023-141817	05/01/2023	01/12/2023	Standard Class	MB-18085	Mick Brown	Consumer	United States	Philadelphia	Pennsylvania	19143	East
6	US-2023-167199	06/01/2023	01/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderson	Kentucky	42420	South
7	US-2023-167199	06/01/2023	01/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderson	Kentucky	42420	South
8	US-2023-106054	06/01/2023	01/07/2023	First Class	JO-15145	Jack O'Briant	Corporate	United States	Athens	Georgia	30605	South
9	US-2023-167199	06/01/2023	01/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderson	Kentucky	42420	South

Figure 1: CLEAN_DATA Sheet showing dataset structure

The key columns used in this analysis include Order Date, Sub-Category, Region, Segment, Sales, Profit and Discount. Three additional derived columns were created to support analysis – Year, Month and Profit Margin%.

Data Cleaning

A thorough cleaning process was carried out before analysis was carried out on the dataset to ensure data integrity and consistency.

A duplicate check confirmed zero duplicate records in the dataset. A blank cell check across all 21 columns using the COUNTBLANK formula returned zero missing values confirming complete data across all key fields.

However 4,130 rows contained inconsistent date formats stored as text which were identified using COUNTIF error checking and corrected using Excel's Text to Columns function converting all dates to a consistent MDY format.

Three derived columns were created – Year using =YEAR() function, Month using =TEXT() function and Profit Margin% calculated as Profit divided by Sales.

Sales	Quantity	Discount	Profit	Year	Month	Profit Margin%
16.448	2	0.2	5.5512	2023	Jan	34%
3.54	2	0.8	-5.487	2023	Jan	-155%
11.784	3	0.2	4.2717	2023	Jan	36%
272.736	3	0.2	-64.7748	2023	Jan	-24%
19.536	3	0.2	4.884	2023	Jan	25%
2573.82	9	0	746.4078	2023	Jan	29%
5.48	2	0	1.4796	2023	Jan	27%
12.78	3	0	5.2398	2023	Jan	41%
609.98	2	0	274.491	2023	Jan	45%

Figure 2 : Derived Columns – Year, Month, Profit Margin%.

Analysis & Findings

Sub-Category Analysis

Analysis of Sub-Category performance revealed that Chairs generate the highest total sales of \$335,768 but maintain only a 4.71% average profit margin making them the most vulnerable Sub-Category to discounting. Paper generates the highest profit margin while Binders record the most negative profit margin confirming their status as the least profitable Sub-Category.



Figure 3 : Sub-Category Sales Ranking

Regional Analysis

Regional analysis revealed that the West region leads in both total sales and profit confirming its status as the strongest performing region. However despite showing positive total profit the Central region maintains a negative average profit margin of -10% indicating widespread unprofitable transactions masked by a few high value orders.

Year over Year Analysis

This revealed that the East region shows consistent sales growth across all years making it the most promising region for future investment. The South region experienced a significant decline in 2024 dropping from \$103,845 to \$71,359 before recovering in 2025 and continuing to grow in 2026.

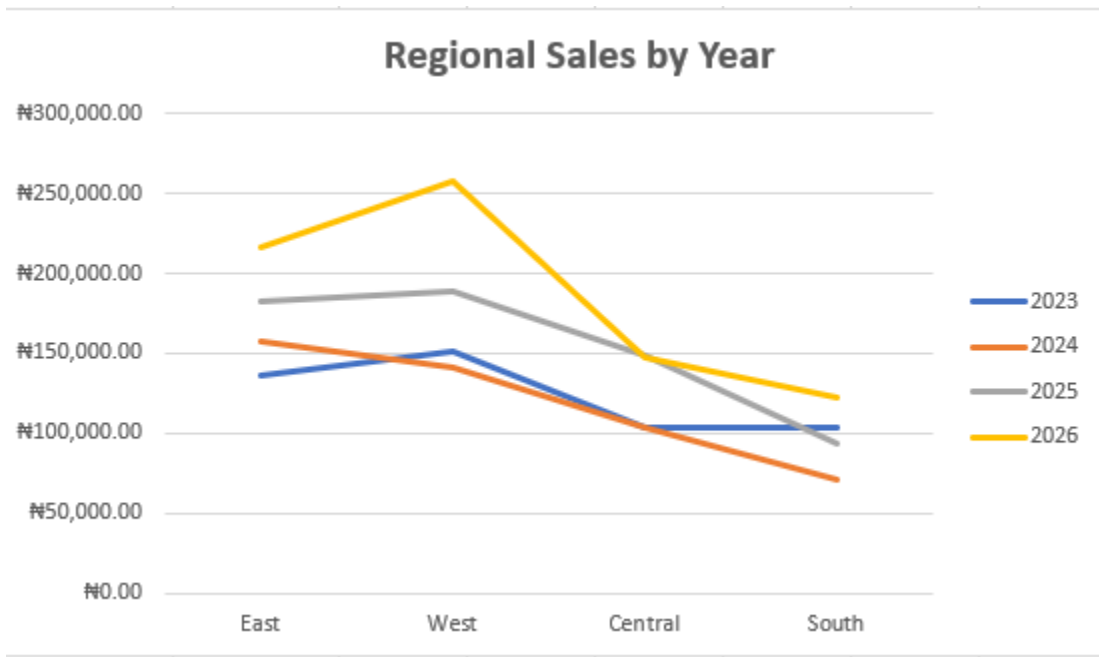


Figure 4 : Regional Sales by Year

Discount Analysis

Analysis of the relationship between discounting and profit margin confirmed that higher discounts consistently lead to lower profit margins. Sub-Categories with average discounts between 16% and 37% showed negative profit margins. Copiers proved to be the notable exception maintaining a strong 32% profit margin despite a 16% average discounting due to their high base value.

SQL Findings

SQL analysis was conducted using SQLiteOnline to validate and extend the Excel findings. Four queries were written and executed against the cleaned dataset.

Query 1 confirmed chairs as the highest selling Sub-Category with dangerously low profit margin consistent with Excel findings.

Query 2 confirmed the West region leads in total sales and profit across all regions.

Query 3 revealed Binders have the highest average discount and the lowest profit margin directly proving the relationship between discounting and profitability.

Query 4 identifies 1,370 loss making orders where discount exceeded 20% with Binders appearing most frequently confirming that aggressive discounting on Binders is the single biggest driver of unprofitable transactions in the business.

```
1 SELECT
2 sub_category,
3 ROUND(SUM(sales), 2) AS Total_Sales,
4 ROUND(SUM(profit), 2) AS Total_Profit,
5 ROUND(AVG(profit/sales)*100, 2) AS Avg_Margin_Pct
6 FROM Superstore_Clean
7 GROUP BY sub_category
8 ORDER BY Total_Sales DESC
9
```

sub_category	Total_Sales	Total_Profit	Avg_Margin_Pct
Chairs	335768.25	27223.53	4.71
Phones	331842.64	45050.83	12.25
Storage	224644.55	21285.11	9.08
Tables	208020.18	-17753.21	-14.02
Binders	207354.88	31426.1	-19.46
Machines	189925.03	3461.98	-6.89

Figure 5 : Sub-Category Sales and Profit Analysis

```

10 SELECT
11 region,
12 ROUND(SUM(sales), 2) AS Total_Sales,
13 ROUND(SUM(profit), 2) AS Total_Profit,
14 ROUND(AVG(profit_margin), 2) AS Profit_Margin
15 FROM Superstore_Clean
16 GROUP BY region
17 ORDER BY Total_Sales DESC
18

```

! region	Total_Sales	Total_Profit	Profit_Margin
West	739813.61	110798.82	22.05
East	691828.17	94883.26	17.08
Central	503170.67	39865.31	-10.46
South	391721.91	46749.43	16.38

Figure 6 : Regional Sales and Profit Analysis

```

19 SELECT
20 sub_category,
21 ROUND(AVG(discount)* 100, 2) AS Discount_Average,
22 ROUND(AVG(profit_margin), 2) AS Profit_Margin_Average
23 FROM Superstore_Clean
24 GROUP BY sub_category
25 ORDER BY Discount_Average DESC
26

```

! sub_category	Discount_Average	Profit_Margin_Average
Binders	36.91	-19.41
Machines	30.43	-6.85
Tables	25.81	-14.05
Bookcases	21.53	-13.51
Chairs	16.92	4.74
Appliances	16.46	-14.87

Figure 7 : Discount vs Profit Margin Analysis

```
27 SELECT
28 COUNT (*) AS Loss_Making_Order
29 FROM Superstore_Clean
30 WHERE Profit < 0
31 AND discount > 0.20

! Loss_Making_Order

1370
```

Figure 8 : Loss Making Orders Analysis

Tableau Dashboard

An interactive Tableau dashboard was created combining four visualizations to provide a comprehensive view of GreenLeaf Markets sales performance. The dashboard includes a geographic map showing profit performance by state, a bar chart ranking Sub-Category sales, a line chart showing regional sales trends and a scatter plot showing the relationship between discounting and profit margins,

The dashboard is published publicly and accessible at:

https://public.tableau.com/views/GreenLeafMarketsSalesAnalysis/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

GreenLeaf Markets - Sales Performance Analysis

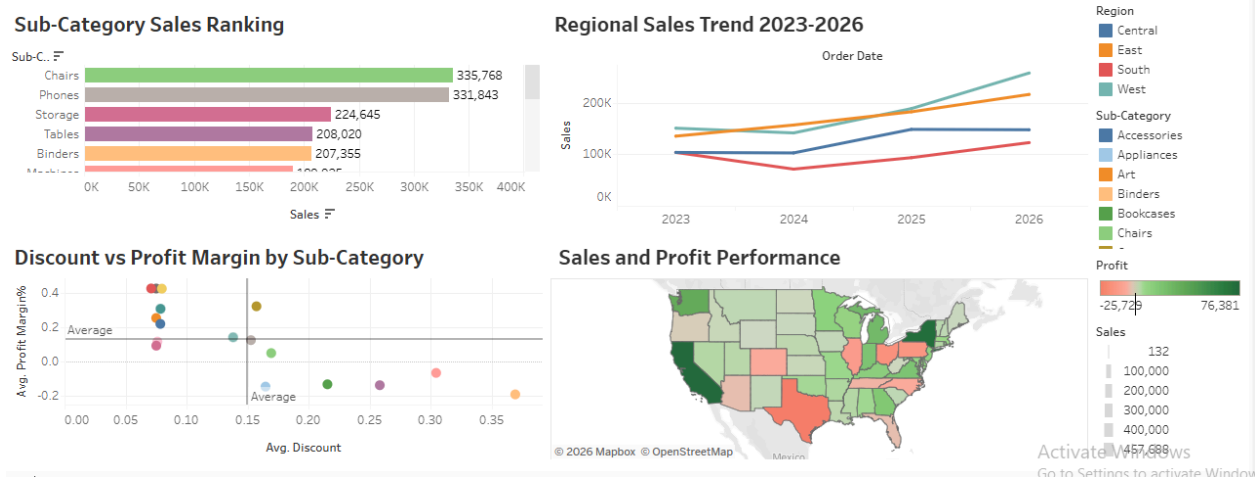


Figure 9 : GreenLeaf Markets Interactive Tableau Dashboard

Recommendations

Cap Discounts on Binders

Analysis shows that Binders have the highest average discount rate and the lowest profit margin. SQL analysis identified them as the most frequent Sub-Category in 1,370 loss making orders. It is recommended that discounts on Binders be capped at a maximum of 15% to prevent further profit erosion.

Conduct a Central Region Pricing Audit

Despite positive total profit the Central region maintains a -10% average profit margin indicating widespread unprofitable transactions masked by a few high value orders. It is recommended that a detailed pricing and discount audit of Central region orders be conducted to identify and address the root cause of widespread losses.

Invest in East Region Growth

The East region shows consistent year over year sales growth making it the most promising region for future investment . It is recommended that additional marketing budget and inventory be allocated to the East region to capitalize on its growth momentum.

Ethics and Limitations

Several ethical considerations and limitations must be acknowledged in this analysis.

Regarding sampling bias – the dataset may not fully represent all customer types or geographic regions equally potentially skewing findings toward certain demographics or locations.

Regarding the impact of recommendations – implementing discount caps based on this analysis could disproportionately affect price sensitive customers who rely on discounts to access products. Any pricing strategy changes should carefully consider the potential impact on customer accessibility and fairness.

Regarding data privacy – the dataset contains customer names and purchase history which raises important privacy considerations. In a real business context this data should be handled in accordance with data protection regulation ensuring customer information is not misused or exposed.

Finally this analysis is limited to sales and profit data and does not account for external factors such as market conditions, competitor pricing or economic changes that may have influenced the findings.

